



**AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH
(AIUB)**

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Summer 2025-2026

Section: [W]

Project Report On

Project Name [Digital Evidence & Forensic Case Management System]

Supervised By

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Obtained Marks for CO2 and CO4 (Description given in the following page)

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =	Evaluation Criteria (CO4)		Total =	
Requirement fulfillment		Design & Innovation			

Validation		Teamwork & Ethics	
Verification		Communication	

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly	Effectively demonstrates the use of validation forms and implements data validation techniques.	Exhibits an exceptional understanding and implementation of validation forms and data

	completely ignored or incorrectly applied.	executed, leading to inadequate handling of data validation.	implemented, resulting in partial or inconsistent handling of data validation.	Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO4: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				

Requirement fulfillment	Fails to demonstrate any attempt to perform or solve system functional requirements. There is no evidence of addressing real-life scenario-based projects or identifying functional requirements for Object-Oriented Programming project development activities.	Demonstrates limited understanding of system functional requirements and real-life scenario-based projects. Attempts to perform or solve functional requirements may be incomplete or inaccurately applied, and there is insufficient consideration given to proper identification of requirements.	Shows a basic understanding of system functional requirements and attempts to perform or solve them. However, the execution may lack depth or thoroughness, and there may be gaps in addressing all aspects of real-life scenario-based projects or properly identifying functional requirements.	Effectively demonstrates the performance and solution of system functional requirements for real-life scenario-based projects. Requirements are identified with clarity and relevance, and the execution is mostly accurate and comprehensive, reflecting a solid understanding of Object-Oriented Programming project development activities.	Exhibits an exceptional understanding of system functional requirements and adeptly performs and solves them for real-life scenario-based projects. Requirements are meticulously identified and executed with thorough attention to detail, showcasing a comprehensive mastery of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms or data validation techniques. There is no evidence of addressing validation requirements or dealing with data validation in their system.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system, although there may be minor issues.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and

					contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any understanding or attempt to verify system data or functional requirements. There is no evidence of addressing verification requirements or considering data flow in their system solution.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts may be incomplete or inaccurately applied, and there is insufficient consideration given to ensuring proper functional requirements and data flow.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may lack thoroughness or precision, and there may be gaps in ensuring proper functional requirements and data flow within the system.	Effectively identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and comprehensive, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

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1. **Introduction**

TraceLock is a desktop-based Digital Evidence and Forensic Case Management System developed to support the organized management of digital investigations. In real-world forensic investigations, large amounts of digital evidence must be associated with specific cases, properly documented, securely maintained, and tracked throughout the investigation process. Managing these records manually can lead to data inconsistency, loss of important information, and difficulties in maintaining accountability.

The TraceLock system provides a centralized platform for managing users, cases, evidence, chain-of-custody records, forensic analysis, evidence files, notifications, and audit logs. The system allows authorized users to organize investigation-related information and maintain a clear record of evidence handling.

The application is developed using C# and .NET 8 Windows Forms to provide a desktop-based graphical user interface. Entity Framework Core 8 is used as the Object-Relational Mapper (ORM) to manage communication between the application and the database. Microsoft SQL Server is used as the relational database for storing and retrieving persistent system data.

A major feature of TraceLock is the management and verification of digital evidence. The system supports chain-of-custody tracking, audit logging, and SHA-256-based evidence integrity verification, helping ensure that important investigation records can be tracked and reviewed. Through input validation and database verification, the system aims to improve the accuracy, integrity, and reliability of digital evidence management.

Overall, TraceLock demonstrates the practical application of Object-Oriented Programming, C# GUI development, database integration, validation, and verification in a realistic digital forensics scenario.

Login Page

This introduces the starting point of the system and shows that TraceLock is a professional desktop application.

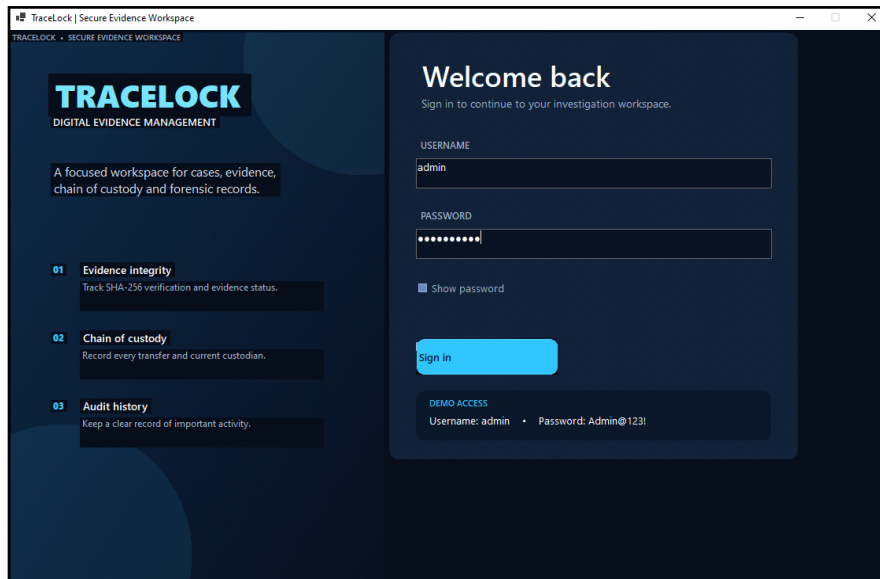


Figure 1.1: TraceLock User Authentication Interface

Dashboard

This gives an immediate overview of the whole system and visually introduces the project.

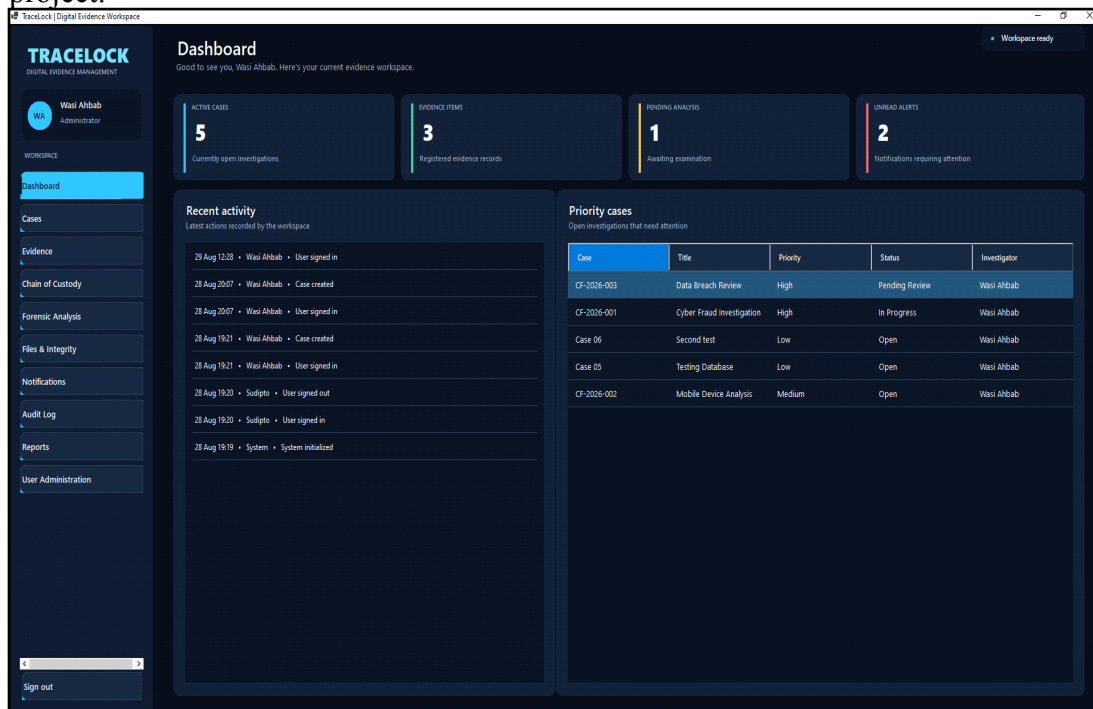


Figure 1.2: TraceLock Digital Evidence Management Dashboard

CHAPTER 02: USER STORY

This chapter presents the main features of TraceLock and explains how different types of users interact with the system. The project is based on a digital evidence and forensic case management scenario where different responsibilities are involved in handling cases, evidence, analysis records, and system activities. Each feature is designed to support a specific part of the investigation and record-management process.

2.1 Feature List

The following features are included in the TraceLock system:

1. User Authentication

The system provides a login process to ensure that only authorized users can access the application. User authentication helps maintain controlled access to the system.

2. Dashboard

The dashboard provides a general overview of the system. It allows users to quickly access important information and navigate to different sections of the application.

3. Case Management

Users can create, update, search, and manage investigation cases. Each case acts as a central record where related information and evidence can be organized.

4. Evidence Management

The system allows users to add and manage digital evidence related to specific cases. Evidence records can be stored and reviewed when required.

5. Chain of Custody

TraceLock maintains a record of evidence handling and transfers. This helps track the movement and responsibility of evidence throughout the investigation process.

6. Forensic Analysis

Users can record analysis information and findings related to digital evidence. These records remain associated with the relevant evidence and case.

7. Evidence File Management

The system allows evidence-related files to be managed alongside their corresponding evidence records, making the files easier to organize and retrieve.

8. Evidence Integrity Verification

TraceLock supports SHA-256-based integrity checking for digital evidence. This allows the system to help verify whether evidence has remained unchanged.

9. Audit Logging

Important activities performed within the system can be recorded in audit logs. These records help maintain accountability and allow actions to be reviewed later.

10. Database Integration

All important system information is stored using Microsoft SQL Server. Entity Framework Core 8 is used to manage the communication between the application and the database.

2.2 User Roles

Different users interact with TraceLock based on their responsibilities within the investigation process.

1. Administrator

The Administrator is responsible for managing the overall system. This role manages user accounts and helps ensure that the application remains properly organized and accessible to authorized users.

2. Investigator

The Investigator is responsible for handling investigation cases. This user can create and manage cases, add related information, and organize evidence associated with an investigation.

3. Evidence Officer

The Evidence Officer is responsible for maintaining evidence records. This includes registering evidence and recording information related to the handling and transfer of evidence.

4. Forensic Analyst

The Forensic Analyst works with digital evidence and records analysis results. This role is responsible for documenting examination findings related to evidence.

5. System Auditor

The System Auditor reviews system records and activities. This role focuses on checking audit information and reviewing actions performed within the system.

2.3 User Stories

The following user stories describe how different users would use the TraceLock system in a practical environment.

User Story 1 — Administrator

As an Administrator, I want to manage user accounts so that authorized users can access and use the TraceLock system.

The administrator can manage user information and monitor access to the application.

User Story 2 — Investigator

As an Investigator, I want to create and manage investigation cases so that all information related to a case can be organized in one place.

The investigator can create cases and maintain relevant case information throughout the investigation.

User Story 3 — Evidence Officer

As an Evidence Officer, I want to register digital evidence and maintain its chain of custody so that the handling history of the evidence can be properly recorded.

The evidence officer can associate evidence with cases and record custody-related activities.

User Story 4 — Forensic Analyst

As a Forensic Analyst, I want to record analysis findings for digital evidence so that examination results can be stored and reviewed as part of the investigation.

The forensic analyst can document the results of forensic examination within the system.

User Story 5 — System Auditor

As a System Auditor, I want to review system activities and audit records so that important actions performed within the system can be monitored and verified.

The auditor can review recorded activities to support accountability and record verification.

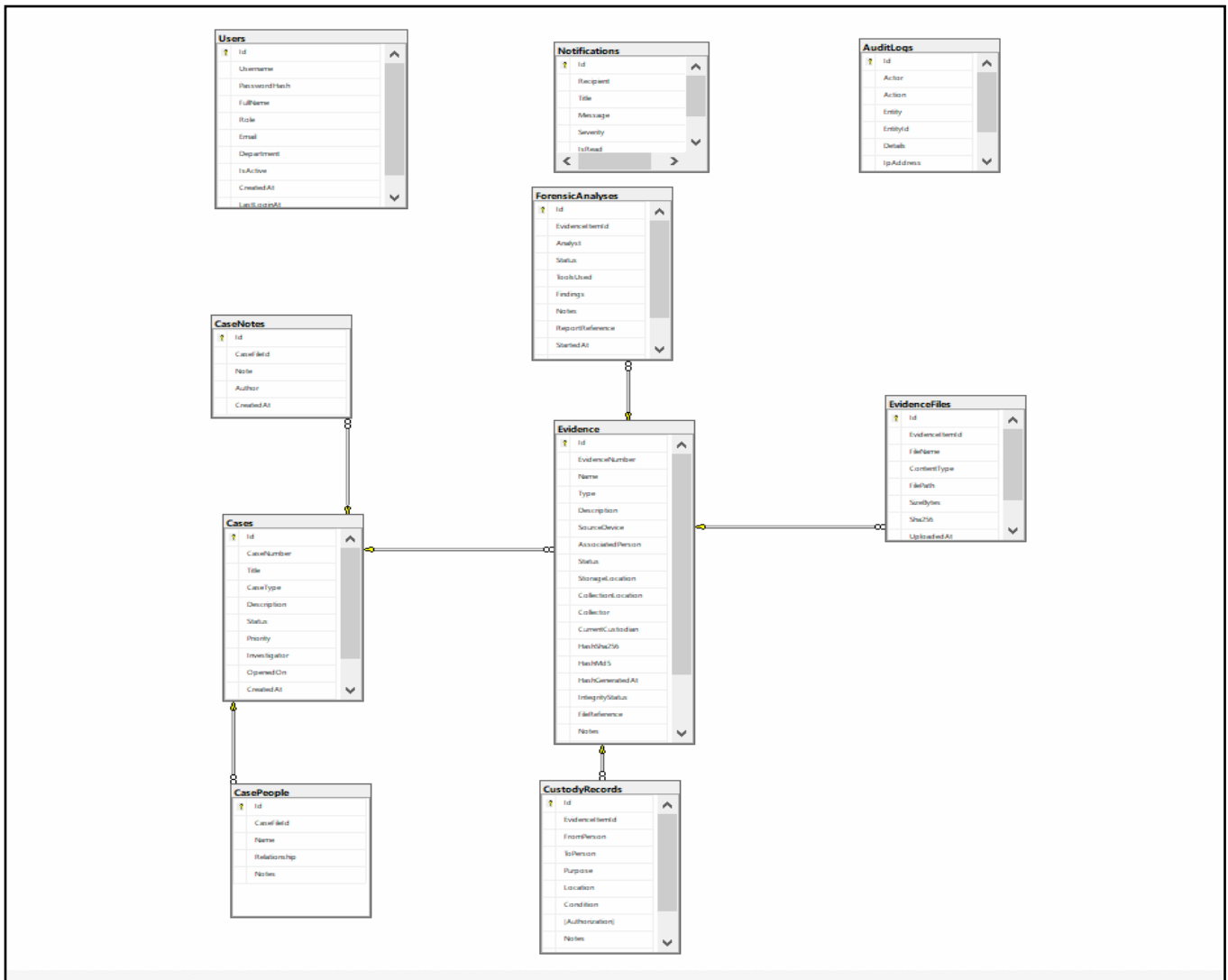
CHAPTER 03a: ER DIAGRAM

Along with diagram description

3a.1 Normalization

Till 3NF

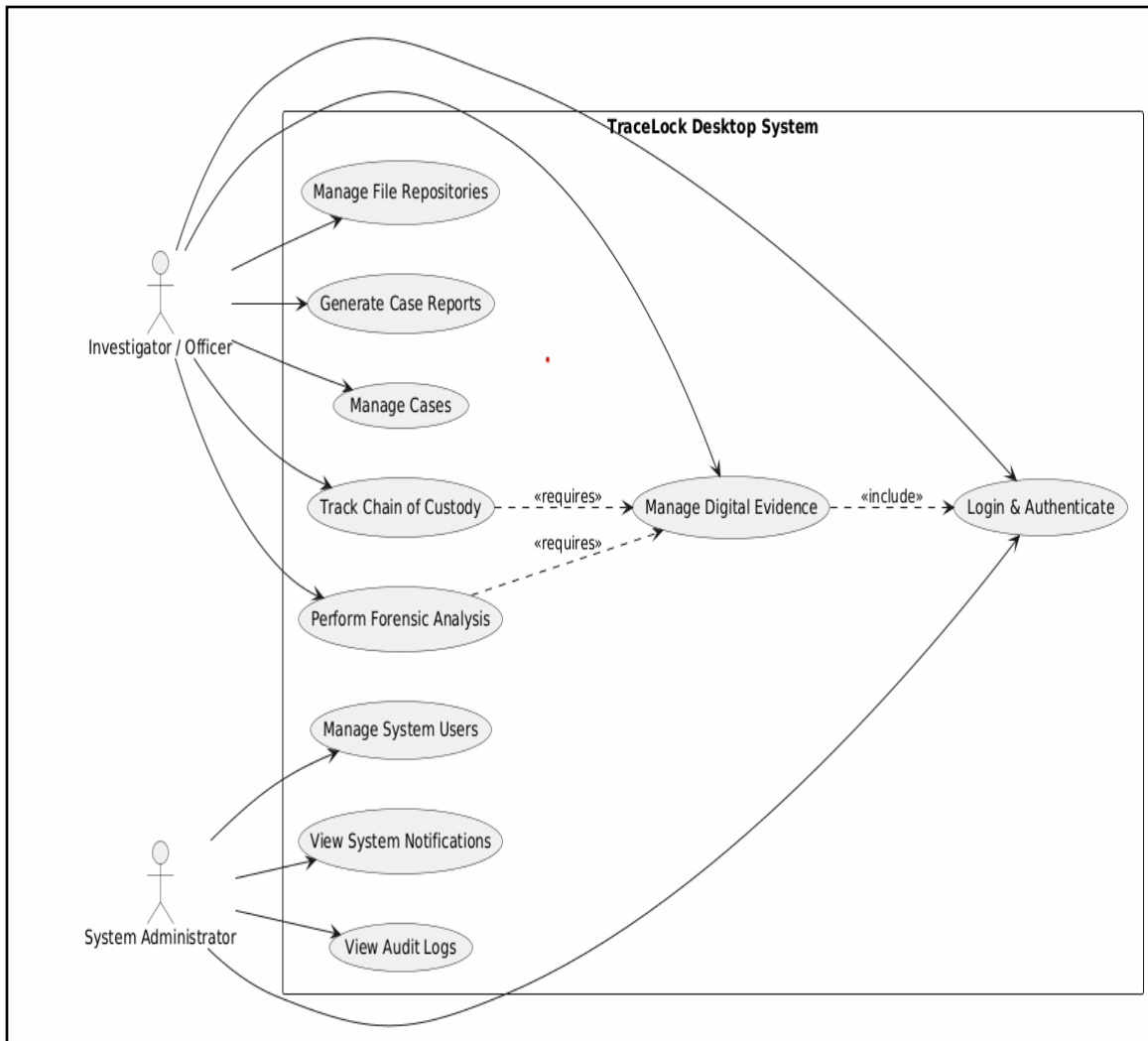
3a.2 Finalization



Chapter 03b: UML Diagram

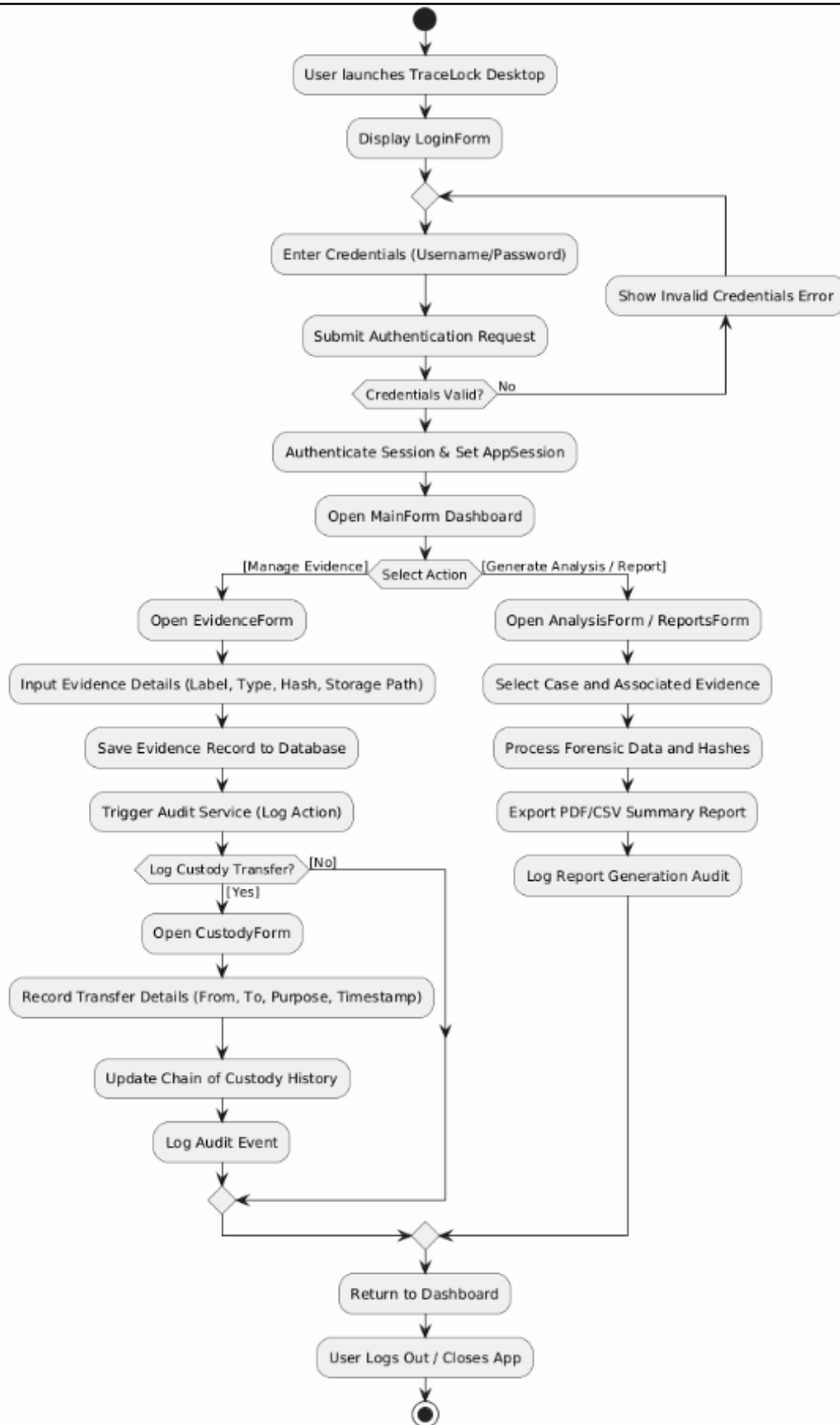
3b.1 Use Case Diagram

Along with Diagram description



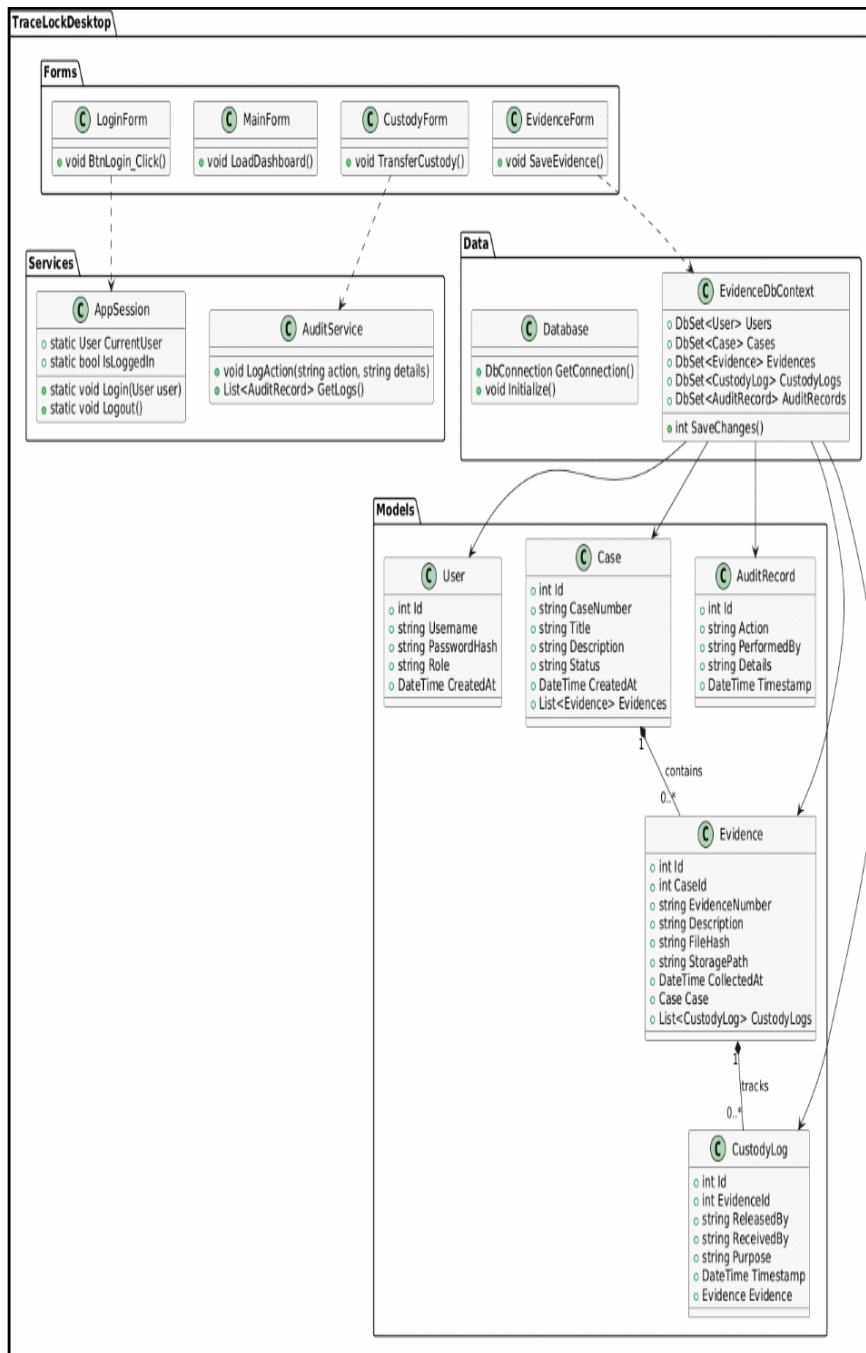
3b.2 Activity Diagram

Along with diagram description



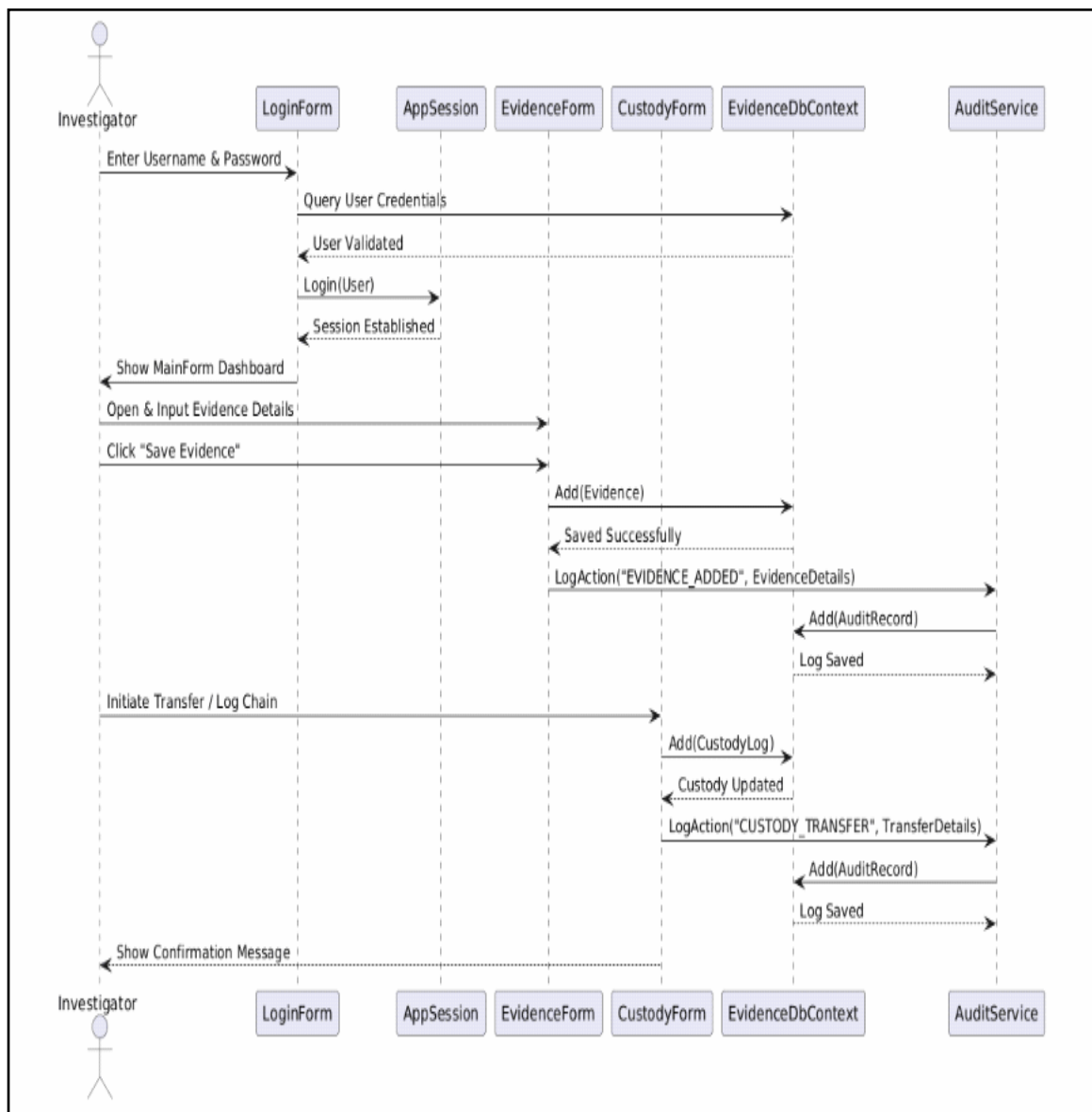
3b.3 Class Diagram

Along with diagram description



3b.4 Sequence Diagram

Along with diagram description



04. Screenshots

Case Management — Displays the creation, organization, and management of digital forensic investigation cases.

The screenshot displays the 'Cases | TraceLock' application window. At the top, there is a search bar and three buttons: 'New case' (highlighted in blue), 'Edit selected', and 'Close case'. Below this is a table listing several cases. The table has columns for Case ID, Title, Type, Priority, Status, Investigator, and Opened date. Below the table, a 'New case' modal form is open, showing fields for Case number, Title, Case type (set to Cybercrime), Status (set to Open), Priority (set to Low), Investigator (set to Wasi Ahbab), Opened date (set to 8/29/2026), and a Description field.

Case	Title	Type	Priority	Status	Investigator	Opened
Case 06	Second test	Cybercrime	Low	Open	Wasi Ahbab	8/28/2026
Case 05	Testing Database	Incident Response	Low	Open	Wasi Ahbab	8/28/2026
CF-2026-003	Data Breach Review	Incident Response	High	Pending Review	Wasi Ahbab	8/25/2026
CF-2026-002	Mobile Device Analysis	Device Examination	Medium	Open	Wasi Ahbab	8/21/2026
CF-2026-001	Cyber Fraud Investigat...	Cybercrime	High	In Progress	Wasi Ahbab	8/16/2026

New case	
Case number	
Title	
Case type	Cybercrime
Status	Open
Priority	Low
Investigator	Wasi Ahbab
Opened	8/29/2026
Description	

Evidence Management — Provides an overview of digital evidence records associated with forensic cases.

Evidence Registry TraceLock								
Search		Add evidence		Edit selected		Verify integrity		Files
Evidence	Name	Type	Case	Status	Custodian	Integrity	SHA256	Collected
EV-003	Network log arch...	Network capture	CF-2026-003	Reviewed	Forensic Analyst	Not verified		8/26/2026
EV-002	Mobile phone ext...	Mobile phone	CF-2026-002	Received	Evidence Custodi...	Not verified		8/22/2026
EV-001	Laptop disk image	Hard drive	CF-2026-001	In Examination	Wasi Ahbab	Not verified		8/17/2026

Add Evidence — Demonstrates the process of registering and adding new digital evidence to a case.

Evidence number	EV-004
Name	Dexter Morgan
Type	Network capture
Status	Received
Case	CF-2026-003 — Data Breach Review
Source device	Blackberry
Associated person	Debra Morgan
Storage location	QV-004 Locker
Collection location	Beach side
Collector	Masuka
Current custodian	Wasi Dexter Morgan
Collected on	8/29/2026
Description	A blackberry phone with multiple broken buttons & sealed camera system with a sim card inserted.
Notes	Could be the main trail of the killer

Save evidence
Cancel

Chain of Custody — Records and tracks the complete history and transfer of evidence to maintain its integrity and accountability.

Chain of Custody | TraceLock

Record transfer

Refresh

Evidence	From	To	Purpose	Location	Condition	Transferred	Recorded
EV-001	Sudipto Mondal	Dexter Morgan	To lab check	Laboratory	Sealed / intact	8/29/2026 1:47 PM	8/29/2026 1:49 PM

Forensic Analysis — Shows the tools and information used to examine and document forensic evidence.

New forensic analysis

Evidence

EV-004 — Dexter Morgan

Analyst

Dexter Morgan

Status

In Progress

Tools used

Autopsy Machine

Started

29 Aug 2026 13:50

Report reference

D*****g

Findings

Few destroyed parts

Notes

Next tests findings

Save analysis

Cancel

Integrity Verification — Verifies the authenticity and integrity of evidence using cryptographic hash values.

Evidence Files & Integrity | TraceLock

EVIDENCE

All stored evidence files

Attach file

Verify selected

Open selected

Evidence	File	Size	Hash	UploadedBy	Uploaded
EV-004	Capture.PNG	3482274	2FCBF5C805878EE44160C223F005F310D31A463EC9733E40312FAEBE1F03A474	Dexter Morgan	8/29/2026 1:53 PM

Integrity check

i

SHA-256:
2FCBF5C805878EE44160C223F005F310D31A463EC9733E40312FAEBE1F03A474

Result: VERIFIED

OK

Audit History / Reports — Displays system activity records and generated reports for investigation tracking and documentation in real time saving to the using device.

Audit Log | TraceLock

Search

Refresh

Time	Actor	Action	Entity	Id	Details	Source
8/29/2026 1:53 PM	Dexter Morgan	Evidence file integrity c...	EvidenceFile	1	Capture.PNG: verified.	Local desktop
8/29/2026 1:53 PM	Dexter Morgan	Evidence file integrity c...	EvidenceFile	1	Capture.PNG: verified.	Local desktop
8/29/2026 1:53 PM	Dexter Morgan	Evidence file accessed	EvidenceFile	1	Capture.PNG was open...	Local desktop
8/29/2026 1:53 PM	Dexter Morgan	Evidence file attached	EvidenceFile	4	Capture.PNG attached t...	Local desktop
8/29/2026 1:51 PM	Dexter Morgan	Forensic analysis created	ForensicAnalysis	2	Analysis for EV-004 — ...	Local desktop
8/29/2026 1:49 PM	Dexter Morgan	Chain-of-custody transf...	Custody	1	EV-001 transferred fro...	Local desktop
8/29/2026 1:47 PM	Dexter Morgan	User signed in	User	5	forensic signed in to th...	Local desktop
8/29/2026 1:44 PM	Wasi Ahbab	User signed out	User	1	admin closed the works...	Local desktop
8/29/2026 1:43 PM	Wasi Ahbab	Evidence registered	Evidence	4	EV-004 — Dexter Morg...	Local desktop
8/29/2026 12:28 PM	Wasi Ahbab	User signed in	User	1	admin signed in to the ...	Local desktop
8/28/2026 8:07 PM	Wasi Ahbab	Case created	Case	5	Case 06 — Second test.	Local desktop
8/28/2026 8:07 PM	Wasi Ahbab	User signed in	User	1	admin signed in to the ...	Local desktop
8/28/2026 7:21 PM	Wasi Ahbab	Case created	Case	4	Case 05 — Testing Data...	Local desktop
8/28/2026 7:21 PM	Wasi Ahbab	User signed in	User	1	admin signed in to the ...	Local desktop
8/28/2026 7:20 PM	Sudipto	User signed out	User	2	investigator closed the ...	Local desktop

Reports | TraceLock

Cases: 5
Transfers: 1

Evidence: 4
Analyses: 2

Evidence CSV

Custody CSV

Audit CSV

Evidence	Name	Type	Case	Status	Custodian	Integrity	SHA256	Collected
EV-001	Laptop disk im...	Hard drive	CF-2026-001	In Examination	Dexter Morgan	Not verified		8/17/2026
EV-002	Mobile phone ...	Mobile phone	CF-2026-002	Received	Evidence Cust...	Not verified		8/22/2026
EV-003	Network log ar...	Network captu...	CF-2026-003	Reviewed	Forensic Analyst	Not verified		8/26/2026
EV-004	Dexter Morgan	Network captu...	CF-2026-003	Received	Wasi Dexter M...	Verified	2FCBF5C80587...	8/29/2026

A Final Database Preview in Microsoft SQL Server (Audit)

Results		Messages						
	ID	Actor	Action	Entity	EntityId	Details	IpAddress	OccurredAt
1	1	System	System initialized	System		TraceLock desktop evidence workspace initialized ...	Local desktop	2026-08-28 19:19:57.3220182
2	2	Sudipto	User signed in	User	2	investigator signed in to the TraceLock desktop wor...	Local desktop	2026-08-28 19:20:42.0341507
3	3	Sudipto	User signed out	User	2	investigator closed the workspace.	Local desktop	2026-08-28 19:20:51.7051617
4	4	Wasi Abbab	User signed in	User	1	admin signed in to the TraceLock desktop workspa...	Local desktop	2026-08-28 19:21:03.9595316
5	5	Wasi Abbab	Case created	Case	4	Case 05 - Testing Database.	Local desktop	2026-08-28 19:21:30.2018431
6	6	Wasi Abbab	User signed in	User	1	admin signed in to the TraceLock desktop workspa...	Local desktop	2026-08-28 20:07:32.3767927
7	7	Wasi Abbab	Case created	Case	5	Case 06 - Second test.	Local desktop	2026-08-28 20:07:50.8921526
8	8	Wasi Abbab	User signed in	User	1	admin signed in to the TraceLock desktop workspa...	Local desktop	2026-08-29 12:28:44.4070590
9	9	Wasi Abbab	Evidence registered	Evidence	4	EVI-004 - Dexter Morgan.	Local desktop	2026-08-29 13:43:14.3329625
10	10	Wasi Abbab	User signed out	User	1	admin closed the workspace.	Local desktop	2026-08-29 13:44:39.4834862
11	11	Dexter Morgan	User signed in	User	5	forensic signed in to the TraceLock desktop worksp...	Local desktop	2026-08-29 13:47:26.0133978
12	12	Dexter Morgan	Chain-of-custody transfer recorded	Custody	1	EVI-001 transferred from Sudipto Mondal to Dexter M...	Local desktop	2026-08-29 13:49:11.4453367
13	13	Dexter Morgan	Forensic analysis created	ForensicAnalysis	2	Analysis for EVI-004 - Dexter Morgan is In Progress.	Local desktop	2026-08-29 13:51:47.2302659
14	14	Dexter Morgan	Evidence file attached	EvidenceFile	4	Capture PNG attached to EVI-004, SHA-256 recorded.	Local desktop	2026-08-29 13:53:16.7934382
15	15	Dexter Morgan	Evidence file accessed	EvidenceFile	1	Capture PNG was opened.	Local desktop	2026-08-29 13:53:21.2298343
16	16	Dexter Morgan	Evidence file integrity checked	EvidenceFile	1	Capture PNG verified.	Local desktop	2026-08-29 13:53:26.0217151
17	17	Dexter Morgan	Evidence file integrity checked	EvidenceFile	1	Capture PNG verified.	Local desktop	2026-08-29 13:53:52.3498633

Query executed successfully.

(localdb)\MSSQLLocalDB (17,...) WASI\moham (39) TraceLock\DigitalEvidence 0000:00 Row: 1, Col: 1 17 rows

05. Conclusion

TraceLock successfully demonstrates the development of a practical desktop-based Digital Evidence and Forensic Case Management System for organizing and managing digital investigations. The project was designed to address the difficulties of manually handling forensic cases and evidence by providing a centralized system where cases, evidence, evidence files, chain-of-custody records, forensic analysis, notifications, and audit logs can be managed in an organized manner.

One of the main achievements of this project is the practical implementation of C# and Object-Oriented Programming (OOP) in a complete software application. C# was used throughout the system to develop the graphical user interface, application logic, validation, database operations, and core functionalities. OOP principles such as encapsulation, abstraction, inheritance, and polymorphism were applied to structure the application into meaningful classes and objects, allowing different components of TraceLock to have clear responsibilities. This approach made the system more organized, maintainable, and easier to extend.

TraceLock was developed using C# with .NET 8 Windows Forms, while Entity Framework Core 8 was used to connect the object-oriented application with Microsoft SQL Server. This integration demonstrates how C# classes and OOP concepts can be effectively used together with a relational database to create a complete data-driven application.

The project also places strong emphasis on evidence integrity, accountability, and traceability. Chain-of-custody records maintain a history of evidence handling, audit logs provide a record of important system activities, and SHA-256 hashing is used to verify the integrity of digital evidence. Together, these features help ensure that forensic information can be properly recorded, monitored, and reviewed throughout an investigation.

Overall, TraceLock showcases the practical application of C#, Object-Oriented Programming, GUI development, database management, Entity Framework Core, validation, and digital evidence integrity within a realistic digital forensics scenario. Rather than being only a demonstration of individual programming concepts, the project brings these concepts together into a functional and structured software solution. Through TraceLock, the project demonstrates how the principles learned in programming and software development can be applied to solve a real-world problem while maintaining organization, reliability, and accountability in digital evidence management.